

20.1.2023

Unit-5Parallel Processors from client to cloud* Multiprocessors

- * highly available systems
- * A computer system with atleast 2 processors is called multiprocessor.
- * They are highly available because on multiprocessor system with n processors, even if a single processor fails $(n-1)$ processors will continue to provide service.
- * They provide high throughput (no. of tasks executed per unit time)

* Multicore microprocessors (parallel processing)

- * A microprocessor containing multiple computing cores in a single integrated circuit.
- * Multicore processors are shared memory processors (SMPs)
- * They usually share single physical address space.

* Difficulties in creating parallel processing programs:
[In general for multiprocessor, multicore]

- 1) Scheduling
- 2) Partitioning work into parallel pieces.
- 3) Balancing the load equally between processors.
- 4) Time to synchronise.
- 5) Overhead for communication between processors.

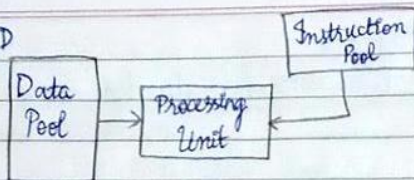
* Cluster - A no. of computers interconnected via communication channel.

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* Flynn's Classification → 4 types

- 1) SISD (Single Instruction stream & Single Data stream)
- 2) SIMD (Single Instruction stream & Multiple Data stream)
- 3) MISD (Multiple Instruction stream & Single Data stream)
- 4) MIMD (Multiple Instruction stream & Multiple Data stream)

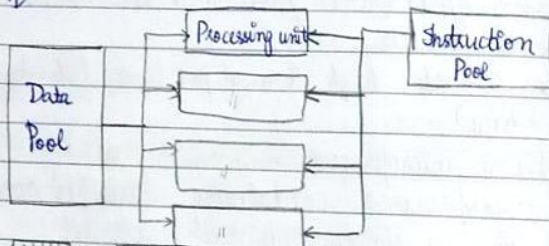
Earlier used SISD



* Advantages

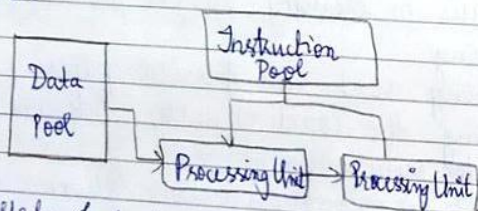
- 1) Very cheap
- 2) Power consumption is less

Widely used SIMD



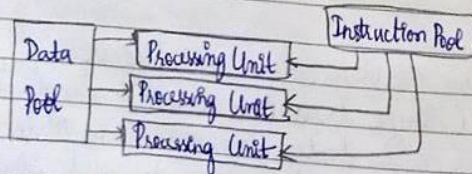
- * Single Instruction can operate on multiple data pieces.
- * Pipelining

Rare 3) MISD



- * Multiple Instruction operating simultaneously on same data
- * Apps: Flight control systems, fault detection system

Widely used 4) MIMD



1) SISD

Single Processor takes data from single address in memory and perform a single instruction on the data at a time. All Single Processors systems are SISD

* Advantages:-

- 1) Low cost
- 2) Low Power Consumption

* Disadvantages:-

- 1) Limited Speed due to single Processors

* 2) Uses:-

- 1) Multicontroller
- 2) Older Mainframes

2) SIMD

* Single Instruction executed on multiple diff pieces of data.

* Instruction can be performed sequentially taking advantage of pipelining or in parallel, using multiple processors.

* Modern GPU's containing vector processors and array processors are SIMD systems.

3) MISD

* Multiple Processors work on the same set of data performing different instructions at the same time.

* Very limited Applications so no commercial availability.

* Uses:-

- 1) Flight Control Systems
- 2) Fault Detection System

4) MIMD

* Multiple Autonomous processors perform operations on diff pieces of data either independently or as part of

shared memory space

* Several diff. instructions can be executed at the same time using diff data streams.

* Ideal for application where multitasking is required.

* Uses -

1) Modern PC's

2) Laptops

3) Smartphones

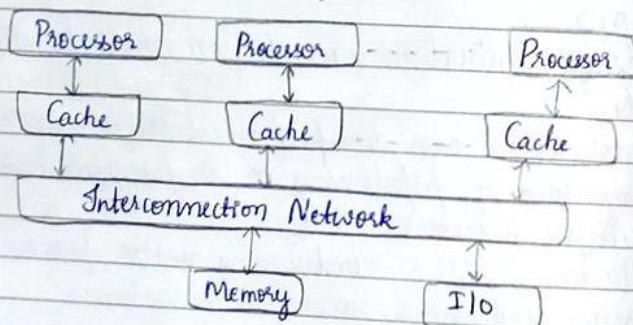
* Disadvantages

1) Complex architecture

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* Multicore and other shared memory multiprocessors

Classic Organization of SMP

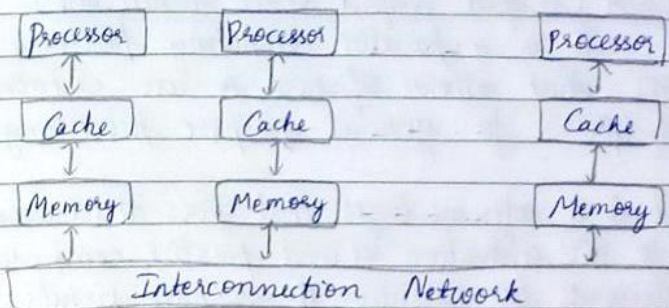


* 2 types of Access:

1) Uniform Memory Access (UMA)

2) Non-Uniform Memory Access (NUMA)

* Clusters, Warehouse Scale Computers and Message Passing Multiprocessors



Message Passing Multiprocessor

* Cloud - collection of multiple servers.

* Warehouse - Very large cluster of computers
scale computers

* GPU's (Graphic Processing Units)

A specialized processor designed to accelerate graphics rendering. (Converting model into 2D/3D image)

* It can process many pieces of data simultaneously making them useful for machine learning, video editing, gaming etc.

* GPGPU - General Purpose GPU

CPU	GPU
1) Less Computing cores	1) More no. of computing cores.
2) Bigger Cache, more complex control units & ALUs	2) Smaller Cache, less complex control units and ALUs.
3) High clock rate and low latency serial processing. (Fetch-decode-execute)	3) Large Parallel Computing Workloads.
4) Complicated Instruction Cycles.	4) Simple Instruction Cycles.

GP GPU - similar to CPU

classmate

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* CUDA (Compute Unified Device Architecture)

It is a parallel computing platform and API that allows software to use certain type of GPU's for general purpose processing.

* It is a software layer that gives direct access to GPU instruction set and parallel computational elements for execution of compute kernels.